



SMART INDIA HACKATHON 2026



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AI-POWERED AUTONOMOUS SEARCH & RESCUE DRONE

Problem Statement ID –26177

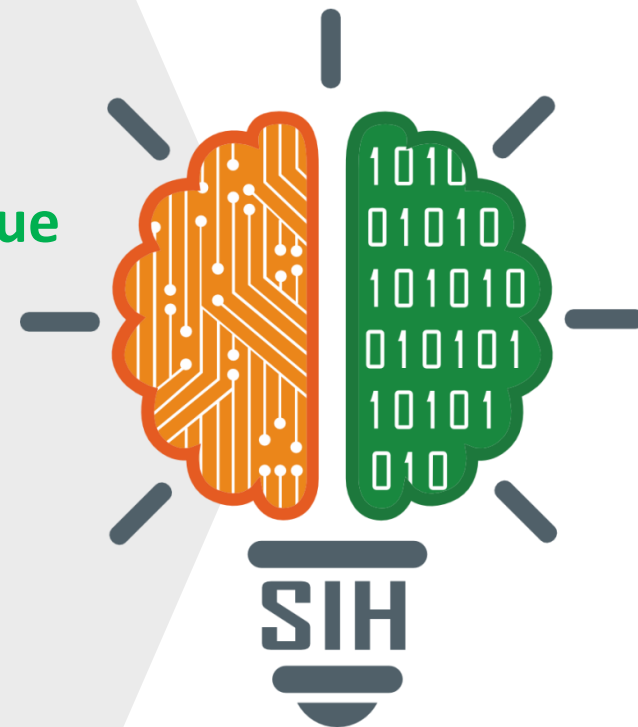
Problem Statement Title- AI-Powered Autonomous Search & Rescue Drone

Theme- Robotics and Drones

PS Category- Hardware

Team ID- 156588

Team Name- GARUDA_001



Problem in Hand

Dangerous Rescue Operations:

Disaster zones may contain unstable structures, floods, fire, debris and other hazards, putting rescuers at risk,

Delayed Victim Discovery

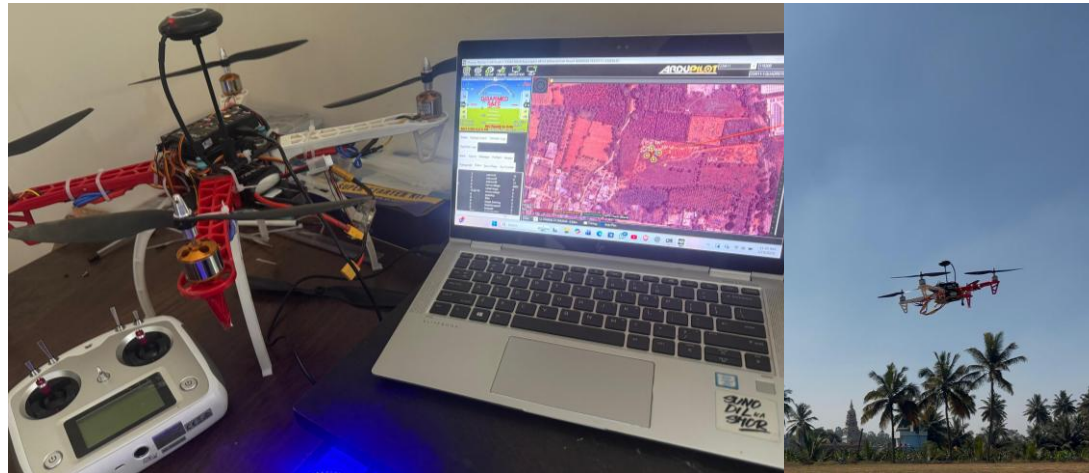
Manual ground-based searching can be slow, especially across large. Damaged or inaccessible areas.

Limited Situational Awareness

Rescue teams may not have a real-time view Of survivors, hazards and safe access routes.

Connectivity Challenges

Disaster areas may have weak or unavailable network connectivity. making cloud-dependent systems unreliable.



Why We Stand Out

On-Device AI (NVIDIA Jetson): Real-time survivor & hazard detection without cloud dependency.

Multi-Sensor Intelligence: Combines RGB, thermal. GPS & IMU data for reliable detection and localization.

Autonomous & GPS-Denied Operation: uses SLAM and obstacle avoidance to navigate in challenging environments.

Actionable Rescue Intelligence: Geo-tagged maps, alerts and prioritized rescue recommendations, and safe access & short routes for rescue teams.

Offline Resilience: Continues critical sensing and On-Device AI operations even when network connectivity is unavailable.

Key Features

On-Device AI Inference

Detects people and signs of human presence using advanced computer vision models.

Thermal Detection

Helps identify human heat signatures in low-visibility conditions (night, smoke, etc.).

Hazard Detection

Identifies fire, smoke. floodwaters, debris, damaged structures and other threats.

Geo-Tagged Mapping

Maps survivor locations, hazards and surveyed areas.

Autonomous Navigation

Supports GPS and GPS-denied environments using SLAM and **obstacle avoidance**.

Emergency Alerting

Provides prioritized alerts and rescue recommendations to responders through **On-Device NVIDIA Jetson**.

Command Center Dashboard

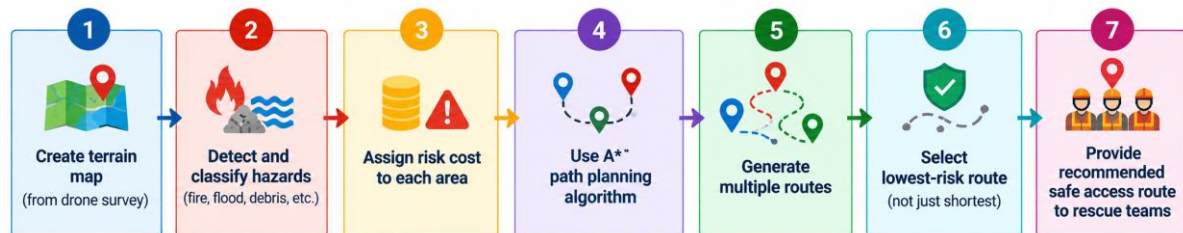
Displays drone status, **detected survivors**, hazards and mission information in real time.

"Let the drone enter the danger zone before the rescuer does."

GARUDA TECHNOLOGY STACK

Hardware Components	Software Components
 Quadcopter Drone Platform Existing frame with motors, ESC and battery	 Python AI and control software
 Flight Controller (Pixhawk) Stable and autonomous flight control	 OpenCV Image processing and computer vision
 RGB Camera & Thermal Camera Visual detection & Detects human heat signatures	 YOLO (Ultralytics) Survivor and hazard detection
 TF-Luna LiDAR Range Finder Altitude estimation, obstacle detection, and safer navigation	 TensorFlow Model training and inference
 GPS Module (Ublox NEO 7M with Compass) Positioning and geotagging	 Mission Planner (MAVLink) Drone communication and mission control
 Edge AI Computer (NVIDIA Jetson) On-device AI processing	 Web Technologies (React + Flask) Rescue command center dashboard
 Real-Time Communication Uses Wi-Fi and transmitter to ground receiver live video, GPS coordinates, and alerts to rescue team.	 MySQL Stores survivor/hazard detections, GPS coordinates, timestamps, risk levels, & data.

How It Works



Risk-aware path planning helps rescue teams reach survivors safer and faster.

FEASIBILITY ANALYSIS

Can we build it with current resources?
YES



Technical Feasibility

All required components (drone frame, cameras, flight controller, Raspberry Pi, GPS, etc.) are commercially available and well-documented.



Economic Feasibility

Estimated prototype cost is within student project budget using off-the-shelf components and open-source software.



Operational Feasibility

Can be operated and tested by a small team with basic training. Suitable for field testing in controlled environments (e.g., campus, rural areas).



Implementation Feasibility

Can be developed in phases using existing tools and open-source frameworks. A working prototype is achievable within the hackathon/six-month timeline.



Scalability & Future Viability

Can be scaled with better sensors, longer flight time and integration with command center systems.

Challenges



Complex terrain & GPS-denied environments



Adverse weather conditions (rain, high wind, low visibility)



Limited flight time (battery constraint)



False positives/negatives in AI detection



No or limited network connectivity

Impacts

Difficult navigation and localization

Affects flight stability and sensor performance

Reduces coverage area per mission

May miss survivors or incorrectly identify objects

Cannot rely on cloud services or real-time communication

Strategy to overcome

Use multi-sensor fusion (RGB + thermal + IMU) and SLAM-based navigation. Implement robust obstacle avoidance and terrain-aware path planning.

Use weather-resistant frame, protective casings and conformal coating. Plan missions within safe weather limits and monitor real-time conditions.

Optimize power consumption (lightweight design, efficient flight path, selective sensing) and plan multi-drone operation for larger coverage.

Train AI models on diverse datasets (day/night, smoke, fire, rubble). Use thermal imaging to complement RGB detection and reduce false alarms.

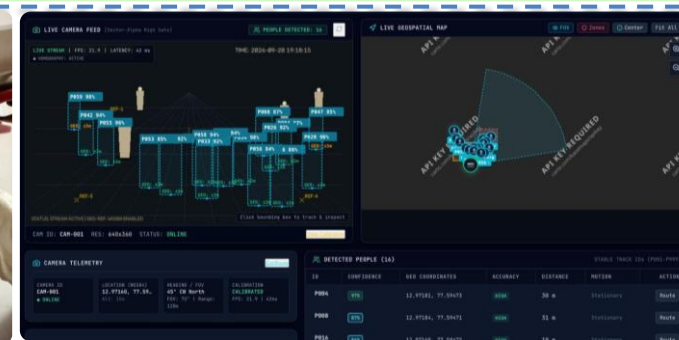
Perform on-device AI processing and store data locally. Use mesh radio / LoRa for short-range communication and sync data when connectivity is available.



Working Prototype with Ground Control Setup



Onboard RGB + Thermal Camera



Garuda Command Center



Human Detection



OVERALL VIABILITY

The project is technically feasible, economically viable, and practically implementable using available components and open-source tools. With a phased approach and proper risk mitigation, GARUDA can evolve into a real-world solution for disaster response, creating meaningful impact and saving lives.

Impact on Stakeholders

Rescue Teams (NDRF/SDRF/Local Units)	Get real-time aerial information on survivors, hazards and safe access routes before entering the disaster area.
Disaster Management Agencies	Gain a live overview of affected zones for better planning, coordination and resource allocation.
First Responders (Police, Medical Teams)	Reduce exposure to dangerous environments by remotely identifying risks such as unstable structures, flood zones.
Affected Communities	Higher chances of being located and assisted quickly, especially in remote or inaccessible areas.
Remote & Inaccessible Areas	Enables aerial assessment where ground-based inspection is difficult due to terrain, floods, landslides.

Benefits of OUR Solution

SOCIAL BENEFITS

- Faster survivor discovery
- Improved responder safety
- Better rescue prioritization
- Inclusive support for vulnerable and remote communities

ECONOMIC BENEFITS

- Reduces search time and manpower
- Reusable platform across multiple missions
- Lower operational costs with on-device processing
- Efficient use of resources

ENVIRONMENTAL BENEFITS

- Battery-powered operation with minimal direct emissions
- Reduces unnecessary ground movement and disturbance
- Supports monitoring of natural hazards (floods, landslides)
- Useful across multiple disaster types

Saving more lives through faster, safer, cost-effective, and sustainable disaster response.

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- **AI-Powered detection** using (RGB + Thermal)
- **Autonomous navigation** & offline-capable
- **Modular and scalable design**

STRENGTHS

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- **Limited flight time** due to limited battery time.
- **High initial hardware cost**
- **Requires technical expertise** of the team.

WEAKNESS

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- **Growing demand** for disaster-response drones
- Government and agency integration
- Multi-drone & **global deployment**

OPPORTUNITIES

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- **Drone regulations** & airspace restrictions
- **Extreme weather** and GPS-denied conditions
- Cybersecurity and **technology competition**

THREATS

A small drone can make a big difference.

References

1. **Qualcomm Flight RB5 5G Platform** — On-device AI, computer vision, autonomous drones, multi-camera processing and edge computing.
[Qualcomm Flight RB5](#)
2. **Ultralytics YOLO Documentation** — Real-time object detection, tracking and edge deployment for survivor/hazard detection.
[Ultralytics YOLO Docs](#)
3. **ROS 2 Documentation** — Robotics middleware for integrating sensors, AI, navigation and drone modules.
4. **MAVLink / QGroundControl** — Drone communication, mission planning, telemetry, mapping and autonomous flight management. QGroundControl supports PX4/ArduPilot vehicles and MAVLink-based communication.
[QGroundControl Documentation](#)
5. **OpenCV** — Computer-vision processing and image/video analysis for the perception pipeline.
6. **Qualcomm Flight RB5 Software & Developer Resources** — Reference software and development resources for building AI-enabled autonomous drone applications.
[Qualcomm Flight RB5 Software](#)

Feature	DJI Matrice 4T	Skydio X10	Flyability Elios 3	Traditional Rescue Drone	GARUDA
Risk-Aware Access Route	● Not core feature	● Not core feature	● Not core feature	● No	● A + hazard cost map*
Multi-Hazard Detection	● Primarily aerial sensing/inspection	● Strong visual/thermal awareness	● Inspection-focused	● Mostly manual	● Person + fire + flood + debris + structural hazards
Offline Edge AI	● Mission-dependent	● Some workflows connectivity-dependent	● Local processing	● Usually limited	● On-device AI + local storage
Prioritized Rescue Alerts	● Command-center integration	● Public-safety workflows	● Inspection reporting	● Manual	● AI-prioritized alerts
Real-Time Disaster Intelligence Map	● Mapping capabilities	● 2D/3D mapping	● 3D LiDAR mapping	● Basic mapping	● Survivors + hazards + risk information
End-to-End Rescue Intelligence	● Aerial sensing platform	● Autonomous public-safety drone	● Specialized inspection drone	● Basic aerial observation	● Detect → Locate → Map → Assess → Recommend → Alert



GARUDA SUPPORTING DOCS

- GITHUB LINK → [Click here](#)
- REPORT LINK → [Click here](#)
- DEMO VIDEO LINK → [Click here](#)

Our solution combines established drone, AI, computer-vision and robotics technologies into a unified disaster-response system.